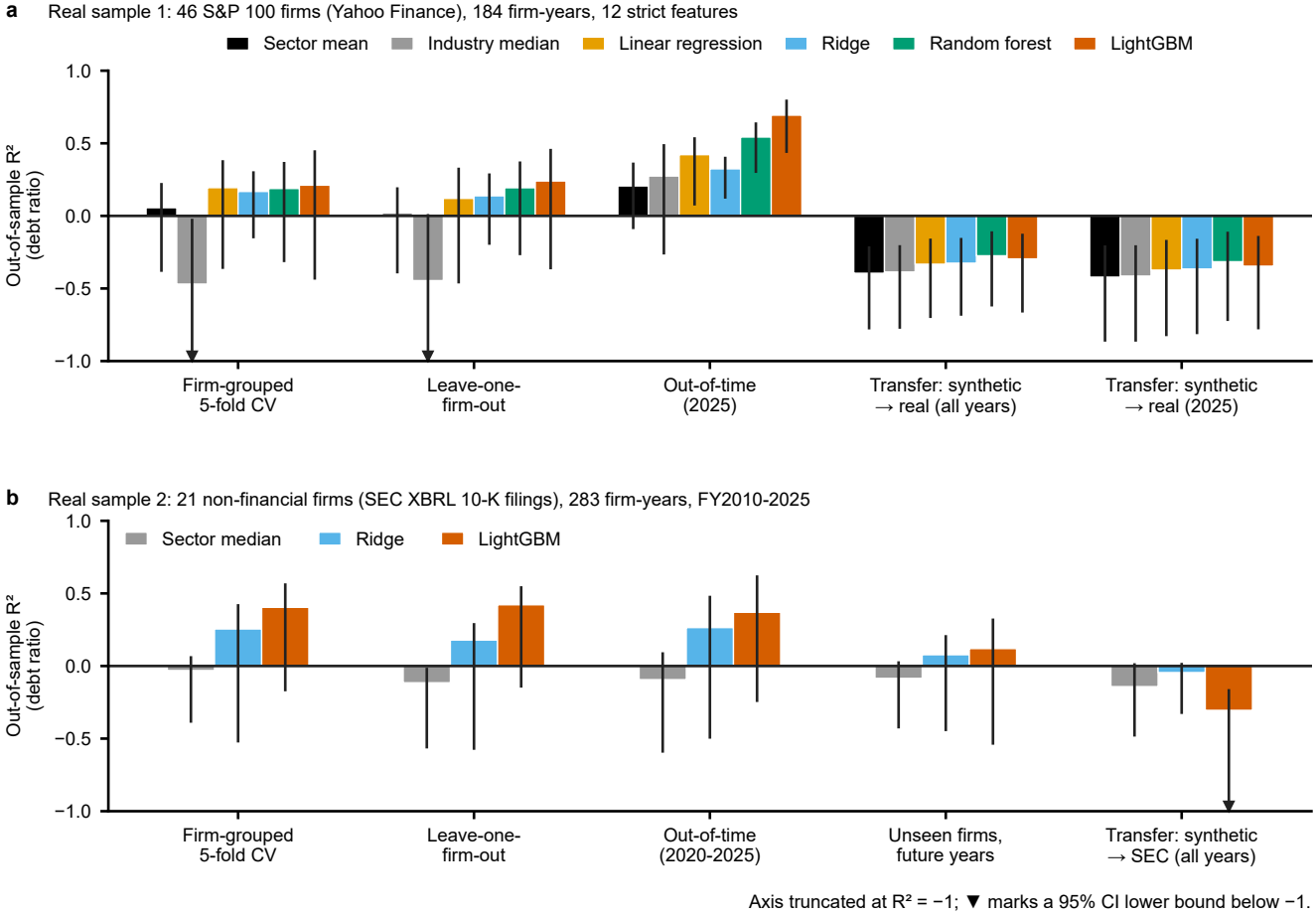
a
Real sample 1: 46 S&P 100 firms (Yahoo Finance), 184 firm-years, 12 strict features
Sector mean
Industry median
Linear regression
Ridge
Random forest
LightGBM
Out-of-sample R²
(debt ratio)
1.0
0.5
0.0
−0.5
−1.0
Firm-grouped
5-fold CV
Leave-one-
firm-out
Out-of-time
(2025)
Transfer: synthetic
→ real (all years)
Transfer: synthetic
→ real (2025)
b
Real sample 2: 21 non-financial firms (SEC XBRL 10-K filings), 283 firm-years, FY2010-2025
Sector median
Ridge
LightGBM
Out-of-sample R²
(debt ratio)
1.0
0.5
0.0
−0.5
−1.0
Firm-grouped
5-fold CV
Leave-one-
firm-out
Out-of-time
(2020-2025)
Unseen firms,
future years
Transfer: synthetic
→ SEC (all years)
Axis truncated at R² = −1; ▼ marks a 95% CI lower bound below −1.